Assignment 01

Importing the Canadian Air Occurrence Record

Teaching Week 04 | Individual work. Submit in R or Python.

Purpose

Retrieve the Canadian air occurrence record, prepare a documented analytical table, and answer one descriptive question. Apply the import, cleaning, summary, and graphics methods from Weeks 01–03. Design and validate a join that adds relevant information while preserving one row per occurrence.

Learning Outcomes

After completing this assignment, you should be able to

- retrieve a published dataset from its official source and record enough provenance for another analyst to repeat the retrieval;
- read a delimited file whose encoding, quoting, and column types are established rather than guessed;
- reduce a wide published extract to a documented analytical table and justify every column you drop;
- separate observations, identifiers, groups, and measures in a record that was not written for analysis;
- compute and interpret centre, spread, quantiles, and missingness for measured counts; and
- join two published tables and show that the join changed only what it should have.

The Dataset

Everyone works on the same record. There is no dataset choice in this assignment, so your reduction and your description are directly comparable with everyone else's.

Air Occurrence Data From January 1995 to Present

Publisher. Transportation Safety Board of Canada, from its Aviation Safety Information System
Official record. `https://www.tsb.gc.ca/eng/stats/aviation/data-5.html`
Licence and format. Open Government Licence – Canada; five comma-separated tables in CSV, published with a data dictionary
Observational unit. One occurrence in the occurrence table. The injuries table contains related detail records; use the dictionary to establish their unit and whether several records refer to the same occurrence.
Variables to locate in the dictionary. occurrence identifier and occurrence number; date and time; occurrence type; occurrence classification; region, province, and country; latitude and longitude; total fatal, serious, minor, none, and unknown injury counts; a free-text summary; and, in the injuries table, the injury category with its own fatal, serious, minor, none, and unknown counts
Use in this assignment. Describe Canadian air occurrence records and compare recorded counts across region, occurrence type, and year. The record does not supply flight-exposure denominators for comparing accident rates.

Retrieving the Data

Download the two data tables and the English data dictionary from the official record; each file is linked from the dataset page. Use a browser download or record the command you used. If a resource is unavailable, retain the error and access time and contact the instructor for a documented course copy; do not silently substitute an unofficial extract.

- **Dataset page.** `https://www.tsb.gc.ca/eng/stats/aviation/data-5.html`
- **Occurrence table.** `ASISdb_MDOTW_VW_OCCURRENCE_PUBLIC.csv`
- **Injuries table.** `ASISdb_MDOTW_VW_INJURIES_PUBLIC.csv`
- **Data dictionary.** `ASISdb-dd.csv`

The catalogue describes monthly updates, so downloads made at different times may differ. Record the access date, byte size, and SHA-256 checksum of each file, including the dictionary. A checksum is a fingerprint of the file's contents, not proof of its retrieval time. Keep the original files privately so your submitted results can be checked against that snapshot. On macOS use `shasum -a 256 filename.csv`; on Linux use `sha256sum filename.csv`. State which snapshot your counts describe.

Data Contract

Reduce the published occurrence extract to one analytical table that satisfies every rule below.

- Retain at least 500 occurrence rows after any filter you apply, and state the filter.

- Keep at least twelve columns, including at least four numeric measures and five categorical variables. Identifiers do not count as measures; record the derivation of any added category such as calendar year.
- Keep at least three grouping variables, each with at least three levels present in the rows you retained.
- Hold one row per occurrence, keyed by the occurrence identifier, with no duplicate key.
- Record the official URL, publisher, licence, retrieval steps, access date, byte size, and SHA-256 checksum for every file.
- Distinguish an empty field from a measured zero, and name the columns that carry each.
- Preserve the dictionary's meaning for every column you keep, and keep identifiers out of numerical summaries.

What the Contract Does Not Settle

Meeting the contract shows that your table is usable. It does not show that your reduction kept the right columns, that your filter is defensible, or that the comparison you chose is fair.

Work Programme

1. Record the Source

State the provenance of each file as specified above. Write one focused descriptive question that the occurrence record can actually answer, and name the observational unit in one sentence. If your question concerns occurrences located in Canada, define that filter explicitly rather than assuming every entry occurred in Canada.

2. Read the File Without Guessing

Write your own import procedure. Establish and document the encoding, delimiter, quoting rules, missing-value codes, column types, and date format for your downloaded snapshot. Preserve the raw files and show that your import handles quoted fields, identifiers, and non-empty values that fail conversion. Report the imported dimensions, any parsing or conversion problems, and how you resolved them using the dictionary.

3. Reduce the Published Extract

Some coded fields have an identifier and English and French display values. Choose the representation your question needs, while retaining keys required to relate tables. Use the data dictionary to decide what survives. Keep the columns your question needs, drop the rest, and give a one-line reason for every group of columns you drop.

4. Audit the Analytical Table

Build a variable table with one row per retained column, giving its type, unit where it has one, missingness, whether it is an identifier, and its analytical role. Say which columns encode an unknown value as a category rather than as an empty field, and how you treat that difference.

5. Describe the Distributions

Report defensible measures of centre, spread, and position for at least three numeric variables, overall and for one grouping variable. Justify each summary against the distribution you observe and say what it hides.

6. Join the Injuries Table

Add at least one useful feature from the injuries table to your occurrence table. Use the dictionary to establish the keys, units, and relationship between the tables, then design and implement an appropriate join. Explain any aggregation or treatment of repeated records that your design requires. Report the expected cardinality, row counts before and after, duplicate or missing keys, and unmatched keys in both directions. Distinguish records outside your chosen subset from records absent from the full source. Your final table must still have one row per retained occurrence. Support any addition of counts with evidence that their units and categories permit it; do not add totals to their components or treat an unmatched value as a measured zero. Explain how your checks establish that the new feature means what you claim it means.

7. Present the Evidence

Present one table and two figures, each with units and a caption that states what it shows. Write two short paragraphs interpreting them against your question, and one paragraph naming a limitation the record itself imposes on your answer. An occurrence count alone does not measure the risk of flying in a region or year. Reporting coverage and investigation status also limit what the record can establish.

Reproducible Workflow

You may complete and submit the assignment in either R or Python. Both paths are marked against the same evidence and the same criteria, and both must run end to end from one documented entry point in your portfolio repository.

Use the import, conversion, summary, and graphics methods taught in Weeks 01–03. For the join, review the Class 07 material on table keys and joins, then establish the observational units and cardinality for these sources. Ask for conceptual clarification if these prerequisites are unclear. Your source must implement the import, selection, conversions, summaries, join, and figures, with checks that support your decisions. Cite any permitted reference you consult.

Record the software and package versions, data paths, and exact command needed to run your work. Generate every reported table, figure, and check from your source. Restart a notebook and run every cell in order before submitting it. Record a seed for any method that depends on randomness.

What to Submit and Where

Keep the work under `assignments/assignment-01/` in a private work repository with instructor access until the deadline has passed and the instructor opens sharing. Commit the finished report, source, and evidence first. In a second commit, record that completed-work commit's full SHA and immutable commit URL in the directory's `README.md`. Push both commits to the default branch by the deadline. A commit cannot contain its own SHA; the marker assesses the snapshot named in the pushed README. The assignment is released on September 29 and is due at 23:59 on October 20, in the course time zone.

Include

- a concise PDF report that answers your descriptive question and states the reduction you made;
- the complete R or Python source plus the generated table, figures, and metrics the report cites;
- the variable audit table and the before-and-after row counts for the join;
- a README carrying the submitted commit SHA and its commit URL, the official data URLs, the retrieval steps, access date, byte size and SHA-256 checksum of each file, the exact execution command, and the environment; and
- links to the original data, with the downloaded snapshot retained privately and available to the marker if requested. Do not put large source extracts in the public portfolio.

A submission is complete only when the required files open and the analysis can be reproduced from the submitted materials. Late work loses 5 percentage points of the available assignment mark for each late calendar day, for up to seven calendar days, subject to the syllabus's accommodation provisions. Cite the dataset and every non-trivial external source, and submit only your own work.

When sharing opens, publish the submitted commit with its existing history for discussion at the next class meeting, if one follows, and in your portfolio. Make corrections in new commits. Do not amend, rebase, squash, force-push, delete, or backdate submitted history. The pushed README record establishes what was submitted and when; Git timestamps alone do not prove timely completion.

Generative AI output is not authorized for this assignment; the syllabus's academic integrity and private-repository exception provisions apply.

Grading

This assignment is worth 7.5 percent of the course grade. The weights below are percentage points of that course grade and sum to the assignment's weight.

Table 1. Assignment 01 marking criteria.

Criterion	Course weight (%)
Provenance, retrieval record, and stated question	1.0
Correct import, explicit typing, and encoding handling	1.5
Column reduction and the variable audit	1.5
Distributional description and its interpretation	1.5
Join correctness and the integrity check around it	1.0
Figures, written argument, and reproducibility	1.0
Total	7.5

How This Differs From the Weekly Labs

The integrated labs are weekly, environmental, and built around a choice among several Nova Scotia and Canadian environmental records. This assignment is individual and uses the same federal aviation safety record for everyone. It brings the import, cleaning, description, and graphics methods together with a documented reduction and a student-designed join. Reusing a lab dataset, a lab question, or a lab report here does not satisfy the assignment.